

# Matrix algebra

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# checklist

- vectors: column vector, row vector, vector addition and subtraction, scalar multiplication, dot product, length (norm), unit vector, angle, linear independence
- matrix: matrix addition and subtraction, scalar multiplication, matrix multiplication (product), matrix transpose, determinant, matrix rank, invertible (non-singular) <sup>squared</sup> matrix, symmetric <sup>must be squared</sup> matrix, orthogonal <sup>squared</sup> matrix, eigendecomposition, SVD, positive-(semi) definite matrix, quadratic forms, trace
- row space, column space, basis, kernel of a linear transformation  
...

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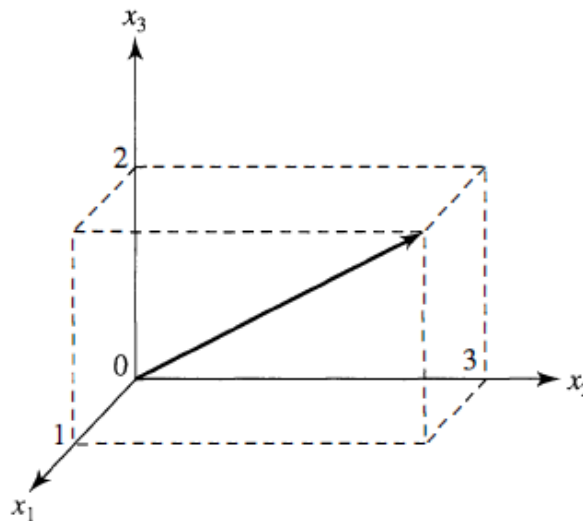
3 Some basics of Matrices

# Vector

An array  $\mathbf{x}$  of  $n$  real numbers  $x_1, \dots, x_n$  is called a vector, and it can be written as

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{or} \quad \mathbf{x}^\top = [x_1, x_2, \dots, x_n]$$

Geometrically, for  $n = 3$ :



# Length, inner product and angle

- The *length* of a vector  $\mathbf{x} = [x_1, x_2, \dots, x_n]^\top$ , with  $n$  components, is defined by

$$\|\mathbf{x}\|_2 := \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}.$$

This is the Euclidean norm.

- For an arbitrary number of dimensions  $n$ , we define the *inner product* of  $\mathbf{x}$  and  $\mathbf{y}$  as

$$\mathbf{x}^\top \mathbf{y} = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

The inner product is denoted by either  $\mathbf{x}^\top \mathbf{y}$  or  $\mathbf{y}^\top \mathbf{x}$ .

- The *angle* of two vectors  $\mathbf{x}$  and  $\mathbf{y}$  satisfies

$$\cos(\theta) = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2} = \frac{\mathbf{x}^\top \mathbf{y}}{\sqrt{\mathbf{x}^\top \mathbf{x}} \sqrt{\mathbf{y}^\top \mathbf{y}}}.$$

We say  $\mathbf{x}$  and  $\mathbf{y}$  are *orthogonal* (or perpendicular) if  $\mathbf{x}^\top \mathbf{y}$  is zero.

# Example

Given the vectors  $\mathbf{x}^\top = [1, 3, 2]$  and  $\mathbf{y}^\top = [-2, 1, -1]$ , find  $3\mathbf{x}$  and  $\mathbf{x} + \mathbf{y}$ .  
Next, determine the length of  $\mathbf{x}$ , the length of  $\mathbf{y}$ , and the angle between  $\mathbf{x}$  and  $\mathbf{y}$ .

First,

$$3\mathbf{x} = 3 \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 9 \\ 6 \end{bmatrix}$$

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 - 2 \\ 3 + 1 \\ 2 - 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$$

Next,  $\mathbf{x}^\top \mathbf{x} = 1^2 + 3^2 + 2^2 = 14$ ,  $\mathbf{y}^\top \mathbf{y} = (-2)^2 + 1^2 + (-1)^2 = 6$ , and  $\mathbf{x}^\top \mathbf{y} = 1(-2) + 3(1) + 2(-1) = -1$ . Therefore,

$$\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^\top \mathbf{x}} = \sqrt{14} = 3.742 \quad \|\mathbf{y}\|_2 = \sqrt{\mathbf{y}^\top \mathbf{y}} = \sqrt{6} = 2.449$$

and

$$\cos(\theta) = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2} = \frac{-1}{3.742 \times 2.449} = -.109$$

so  $\theta = 96.3^\circ$ .

# Linear independence

## Linear independence

A set of vectors  $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$  is said to be *linearly dependent* if there exist constants  $c_1, c_2, \dots, c_k$ , not all zero, such that

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \dots + c_k\mathbf{x}_k = \mathbf{0}$$

Linear dependence implies that at least one vector in the set can be written as a linear combination of the other vectors. Vectors of the same dimension that are not linearly dependent are said to be *linearly independent*.

## Example

Consider the set of vectors

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Setting

$$c_1 \mathbf{x}_1 + c_2 \mathbf{x}_2 + c_3 \mathbf{x}_3 = \mathbf{0}$$

implies that

$$c_1 + c_2 + c_3 = 0$$

$$2c_1 - 2c_3 = 0$$

$$c_1 - c_2 + c_3 = 0$$

with the unique solution  $c_1 = c_2 = c_3 = 0$ . As we cannot find three constants  $c_1, c_2$ , and  $c_3$ , not all zero, such that  $c_1 \mathbf{x}_1 + c_2 \mathbf{x}_2 + c_3 \mathbf{x}_3 = \mathbf{0}$ , the vectors  $\mathbf{x}_1, \mathbf{x}_2$ , and  $\mathbf{x}_3$  are linearly independent.

# Projection of a vector on another vector

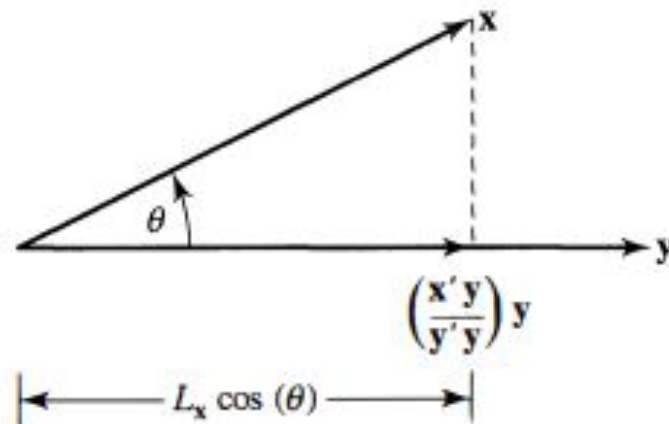
The *projection* of a vector  $\mathbf{x}$  on a vector  $\mathbf{y}$  is

$$\text{Projection of } \mathbf{x} \text{ on } \mathbf{y} = \frac{(\mathbf{x}^\top \mathbf{y})}{\mathbf{y}^\top \mathbf{y}} \mathbf{y} = \frac{(\mathbf{x}^\top \mathbf{y})}{\|\mathbf{y}\|_2} \frac{1}{\|\mathbf{y}\|_2} \mathbf{y}$$

where the vector  $\|\mathbf{y}\|_2^{-1} \mathbf{y}$  has unit length. The length of the projection is

$$\text{Length of projection} = \frac{|\mathbf{x}^\top \mathbf{y}|}{\|\mathbf{y}\|_2} = \|\mathbf{x}\|_2 \left| \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2} \right| = \|\mathbf{x}\|_2 |\cos(\theta)|$$

where  $\theta$  is the angle between  $\mathbf{x}$  and  $\mathbf{y}$ .



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# Matrix multiplication (product)

## Matrix multiplication

The *matrix multiplication*  $\mathbf{AB}$  of an  $m \times n$  matrix  $\mathbf{A} = \{a_{ij}\}$  and an  $n \times k$  matrix  $\mathbf{B} = \{b_{ij}\}$  is the  $m \times k$  matrix  $\mathbf{C}$  whose elements are

$$c_{ij} = \sum_{\ell=1}^n a_{i\ell} b_{\ell j} \quad i = 1, 2, \dots, m \quad j = 1, 2, \dots, k$$

Note that for the product  $\mathbf{AB}$  to be defined, the column dimension of  $\mathbf{A}$  must equal the row dimension of  $\mathbf{B}$ .

Let  $\mathbf{I}$  denote the square matrix with ones on the diagonal and zeros elsewhere, then  $\mathbf{AI} = \mathbf{IA} = \mathbf{A}$ . The matrix  $\mathbf{I}$  acts like 1 in ordinary multiplication ( $1 \cdot a = a \cdot 1 = a$ ), so it is called the *identity matrix*.

# Matrix transpose

- The *transpose* operation  $\mathbf{A}^\top$  of a matrix changes the columns into rows, so that the first column of  $\mathbf{A}$  becomes the first row of  $\mathbf{A}^\top$ , the second column becomes the second row, and so forth. In other words,  $(\mathbf{A}^\top)_{ji} = (\mathbf{A})_{ij}$ .

*reflect rotate in 90° degree*

- Example: If

$$\mathbf{A}_{(2 \times 3)} = \begin{bmatrix} 3 & -1 & 2 \\ 1 & 5 & 4 \end{bmatrix} \quad (\mathbf{A}^\top)_{ji} = A_{ij}$$

$$\mathbf{A} = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{pmatrix} \text{GR}^{m \times n}$$

$$\mathbf{A}^\top = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{pmatrix} \text{GR}^{n \times m}$$

$$\mathbf{A}^\top_{(3 \times 2)} = \begin{bmatrix} 3 & 1 \\ -1 & 5 \\ 2 & 4 \end{bmatrix}$$

*$A_{31} = (\mathbf{A}^\top)_{13}$*

then

# Properties of matrix multiplication

$x \in \left( \begin{array}{c} \mathbb{R}^{m \times p} \text{ image data.} \\ \mathbb{R}^{p \times p} \end{array} \right)$  stored in a matrix.  $\xrightarrow{\text{implemented.}}$   $\text{PCA}$   $\rightarrow$   $\text{TS}$ .

For all matrices  $\mathbf{A}$ ,  $\mathbf{B}$ , and  $\mathbf{C}$  (of dimensions such that the indicated products are defined) and a scalar  $c$ , verify:  $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$ : prove  $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$ . *ij entry is the same for verification.*

- $c(\mathbf{AB}) = (c\mathbf{A})\mathbf{B}$
- $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$
- $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$
- $(\mathbf{B} + \mathbf{C})\mathbf{A} = \mathbf{BA} + \mathbf{CA}$
- $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$

*def of mult*  
 $\textcircled{1} (\mathbf{AB})^T_{ij}$   
 $\textcircled{2} \mathbf{B}^T \mathbf{A}^T_{ij}$   
 For any  $1 \leq i \leq k, 1 \leq j \leq m$   
 $\textcircled{3} (\mathbf{B}^T \mathbf{A}^T)_{ij} = \sum_{b=1}^n (\mathbf{B}^T)_{ib} (\mathbf{A}^T)_{bj} = \sum_{b=1}^n B_{bi} A_{jb} = \sum_{b=1}^n A_{jb} B_{bi} = (\mathbf{AB})_{ji}$   
 $\textcircled{4} (\mathbf{AB})^T_{ij} = (\mathbf{AB})_{ji}$   
 $\therefore (\mathbf{B}^T \mathbf{A}^T)_{ij} = (\mathbf{AB})^T_{ij}$   
 $\therefore \mathbf{B}^T \mathbf{A}^T = (\mathbf{AB})^T$   
 $(\mathbf{ABC})^T = \mathbf{C}^T \mathbf{B}^T \mathbf{A}^T$

There are several important differences between the algebra of matrices and the algebra of real numbers. Two of these differences are as follows:

# Matrix multiplication

product of two matrices

$$(x^T x)^{-1} x^T y$$

$$A^{m \times n} \cdot B^{n \times k} \rightarrow C^{m \times k}$$

with  $A$  and  $B$  as matrices,  $C$  is the result.

$$C = AB \quad C_{ij} \text{-entry}$$

$$C_{ij} = \sum_k A_{ik} B_{kj}, \quad k=1,2,\dots,n.$$

1. Matrix multiplication is, in general, **not commutative**. That is, in general,  **$AB \neq BA$** . For example

$$\begin{bmatrix} 3 & -1 \\ 4 & 7 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 14 \end{bmatrix}$$

but

$$\begin{bmatrix} 0 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 3 & -1 \\ 4 & 7 \end{bmatrix} \quad \text{not defined}$$

is **not defined**.

$$a \cdot I = I \cdot a = a$$

$a$  is a number

$$AI = IA = A$$

$$I = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$$

# Matrix multiplication

2. Let  $\mathbf{0}$  denote the zero matrix, that is, the matrix with zero for every element. In the algebra of real numbers, if the product of two numbers,  $ab$ , is zero, then  $a = 0$  or  $b = 0$ . In matrix algebra, however, **the product of two nonzero matrices may be the zero matrix**. Hence,

$$\begin{matrix} \mathbf{AB} \\ (m \times n)(n \times k) \end{matrix} = \begin{matrix} \mathbf{0} \\ (m \times k) \end{matrix}$$

does not imply that  $\underbrace{\mathbf{A} = \mathbf{0}}_{\text{each entry is 0}}$  or  $\underbrace{\mathbf{B} = \mathbf{0}}$ . For example,

$$\begin{bmatrix} 3 & 1 & 3 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ -5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

It is true, however, that if either  $\underbrace{\begin{matrix} \mathbf{A} & = & \mathbf{0} \\ (m \times n) & & (m \times n) \end{matrix}}_{\text{or}} \begin{matrix} \mathbf{B} & = & \mathbf{0} \\ (n \times k) & & (n \times k) \end{matrix}$ , then

$$\begin{matrix} \mathbf{AB} \\ (m \times n)(n \times k) \end{matrix} = \begin{matrix} \mathbf{0} \\ (m \times k) \end{matrix}.$$

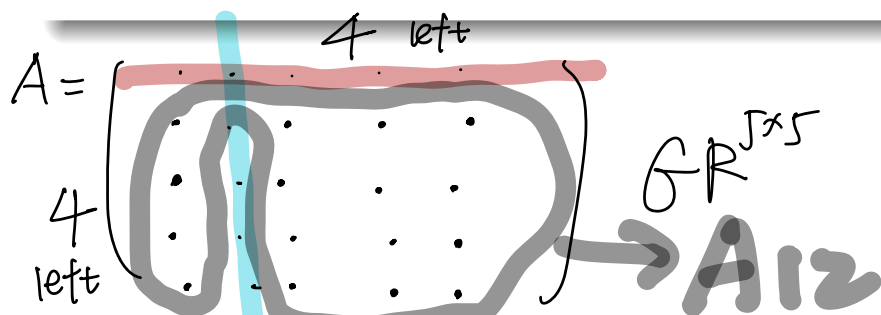
# Determinant *in matrix inverse*

## Determinant *recursively*

The *determinant* of the square  $k \times k$  matrix  $\mathbf{A} = \{a_{ij}\}$ , denoted by  $|\mathbf{A}|$ , is the scalar

$$\begin{aligned} |\mathbf{A}| &= a_{11} && \text{if } k = 1 \\ |\mathbf{A}| &= \sum_{j=1}^k a_{1j} |\mathbf{A}_{1j}| (-1)^{1+j} && \text{if } k > 1 \end{aligned}$$

where  $\mathbf{A}_{1j}$  is the  $(k-1) \times (k-1)$  matrix obtained by deleting the first row and  $j$ th column of  $\mathbf{A}$ . Also,  $|\mathbf{A}| = \sum_{j=1}^k a_{ij} |\mathbf{A}_{ij}| (-1)^{i+j}$ , with the  $i$ th row in place of the first row.



# Example

fast in R for `det()`

- If  $\mathbf{I}$  is the  $k \times k$  identity matrix,  $|\mathbf{I}| = 1$ .



$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22}(-1)^2 + a_{12}a_{21}(-1)^3 = a_{11}a_{22} - a_{12}a_{21}$$



$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} (-1)^2 + a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} (-1)^3 + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} (-1)^4$$

$$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{21}a_{32}a_{13} - a_{31}a_{22}a_{13} - a_{21}a_{12}a_{33} - a_{32}a_{23}a_{11}$$

# Rank important, close to PCA Interpretation, Factor Analysis

- The row rank of a matrix is the maximum number of linearly independent rows, considered as vectors (that is, row vectors). The column rank of a matrix is the rank of its set of columns, considered as vectors.
- The **row rank and the column rank** of a matrix are equal. Thus, the rank of a matrix is either the row rank or the column rank.
- For example, for the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & -1 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow[\text{rank}]{\substack{\text{row reduction} \\ \text{Solve}}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & -3 \\ 0 & 1 & -1 \end{bmatrix}$$

(-2) row<sub>1</sub> + row<sub>2</sub>

the rank is two.

$$\Rightarrow \text{row rank} = 2$$

$$\Rightarrow \text{rank} = 2$$

$$\xrightarrow{-3\text{row}_3 + \text{row}_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

# Matrix inverse

## Matrix inverse

If there exists a matrix  $\mathbf{B}$  such that

$$\underbrace{\mathbf{B} \quad \mathbf{A}}_{(k \times k)(k \times k)} = \underbrace{\mathbf{A} \quad \mathbf{B}}_{(k \times k)(k \times k)} = \underbrace{\mathbf{I}}_{(k \times k)},$$

then  $\mathbf{B}$  is called the inverse of  $\mathbf{A}$  and is denoted by  $\mathbf{A}^{-1}$ .

For a  $k \times k$  matrix  $\mathbf{A}$ :

might have MP-inverse / pseudo inverse not discussed. no squared matrix

the existence of  $\mathbf{A}^{-1}$

$\iff$  <sup>or</sup> the columns (rows) of  $\mathbf{A}$  are linear independent

$\iff$  <sup>or</sup>  $\mathbf{A}$  has full column (row) rank

$\iff$  <sup>or</sup>  $|\mathbf{A}| \neq 0$

# Matrix inverse

- In general,  $\mathbf{A}^{-1}$  has  $(i, j)$ -th entry  $\overset{\text{calculated}}{[(-1)^{j+i} |\mathbf{A}_{ji}| / |\mathbf{A}|]}$ , where  $\mathbf{A}_{ji}$  is the matrix obtained from  $\mathbf{A}$  by deleting the  $j$ th row and  $i$ th column.
- Specifically, the inverse of any  $2 \times 2$  matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

is given by

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}.$$

# $\triangle$ Some as questions Some properties of matrix inverse and determinant

Let  $\mathbf{A}$  and  $\mathbf{B}$  be square matrices of the same dimension, and let the indicated inverses exist. Then the following hold:

- $(\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1}$
- $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$

Let  $\mathbf{A}$  and  $\mathbf{B}$  be  $k \times k$  square matrices.

- $|\mathbf{A}| = |\mathbf{A}^\top|$
- If each element of a row (column) of  $\mathbf{A}$  is zero, then  $|\mathbf{A}| = 0$
- If any two rows (columns) of  $\mathbf{A}$  are identical, then  $|\mathbf{A}| = 0$
- If  $\mathbf{A}$  is nonsingular, then  $|\mathbf{A}| = 1/|\mathbf{A}^{-1}|$ ; that is,  $|\mathbf{A}||\mathbf{A}^{-1}| = 1$ .
- $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$
- $|c\mathbf{A}| = c^k|\mathbf{A}|$ , where  $c$  is a scalar.

$$\det(c\mathbf{A}) = c^k \det(\mathbf{A}) \quad \mathbf{A} \in \mathbb{R}^{k \times k}$$

$$c\mathbf{A} = \begin{bmatrix} cA_{11} & cA_{12} & \dots & cA_{1k} \\ cA_{21} & cA_{22} & \dots & cA_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ cA_{k1} & cA_{k2} & \dots & cA_{kk} \end{bmatrix} \sim \sum A_{i1j1}' A_{i2j2}' \dots A_{ikjk}' \sim \sum c^k \cdot A_{i1j1}' A_{i2j2}' \dots A_{ikjk}'$$

# Diagonal matrix

- The calculations of matrix inverse are usually proceeded by a computer, especially when the dimension is greater than three.
- Diagonal matrices have inverses that are easy to compute. For example,

$$\begin{array}{c} \det = \sum_{i=1}^k a_{ii} \\ \left[ \begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ 0 & a_{22} & 0 & 0 & 0 \\ 0 & 0 & a_{33} & 0 & 0 \\ 0 & 0 & 0 & a_{44} & 0 \\ 0 & 0 & 0 & 0 & a_{55} \end{array} \right] \end{array} \text{ has inverse } \begin{array}{c} \left[ \begin{array}{ccccc} \frac{1}{a_{11}} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{a_{22}} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{a_{33}} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{a_{44}} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{a_{55}} \end{array} \right] \end{array}$$

if all the  $a_{ii} \neq 0$ .

# Symmetric matrix and Orthogonal matrix

$$(A A^T)^T = (A^T)^T A^T = A A^T \Rightarrow A A^T \text{ symmetric.}$$

$$(A+B+C)^T = A^T + B^T + C^T$$

$$X \in \mathbb{R}^{n \times p} \quad S = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^T$$

$$= \frac{1}{n-1} X^T (I - \frac{1}{n} J J^T) X$$

$$S^T = S \quad \text{verified: } S^T = \left( \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^T \right)^T = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^T = S$$

$$(AB)^T = B^T A^T$$

- A square matrix is said to be *symmetric* if

$$A = A^T \text{ or } a_{ij} = a_{ji} \text{ for all } i \text{ and } j.$$

- Another special class of square matrices are the orthogonal matrices, characterized by

eigen vectors

$$Q Q^T = I \quad Q^{-1} Q Q^T = Q^{-1} I \Rightarrow Q^T = Q^{-1}$$

why  $Q$  is invertible for orthogonal matrix  
 $\det(QB) = \det(Q) \det(B)$

$$Q Q^T = Q^T Q = I \text{ or } Q^T = Q^{-1} = \frac{1}{\det(Q) \det(Q^T)} = \frac{1}{(\det(Q))^2}$$

The name derives from the property that if  $Q$  has  $i$ th row  $q_i^T$ , then  $Q Q^T = I$  implies that  $q_i^T q_i = 1$  and  $q_i^T q_j = 0$  for  $i \neq j$ , so the rows have unit length and are mutually perpendicular (orthogonal).

According to the condition  $Q^T Q = I$ , the columns have the same property.

$$\text{If } Q = \begin{pmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_k^T \end{pmatrix}$$

$$Q Q^T = \begin{pmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_k^T \end{pmatrix} (q_1 \quad q_2 \quad \dots \quad q_k)$$

# Eigenvalues and eigenvectors

$$= \begin{bmatrix} p_1^T p_1 & \dots & p_1^T p_k \\ & p_2^T p_2 & \\ & & \ddots \\ p_k^T p_1 & & \dots & p_k^T p_k \end{bmatrix}$$

A square matrix  $\mathbf{A}$  is said to have an eigenvalue  $\lambda$ , with corresponding eigenvector  $\mathbf{x} \neq \mathbf{0}$ , if

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}.$$

$$A(2x) = \lambda(2x) \quad \checkmark$$

$$Ax = (\lambda)x \quad \times$$

$$= \begin{bmatrix} \cancel{0} & \cancel{0} \\ 0 & 0 \end{bmatrix}$$

- Let  $\mathbf{A}$  be a  $k \times k$  square matrix and  $\mathbf{I}$  be the  $k \times k$  identity matrix.

Then the scalars  $\lambda_1, \lambda_2, \dots, \lambda_k$  satisfying the polynomial equation

$|\mathbf{A} - \lambda\mathbf{I}| = 0$  <sup>unit function</sup> are the eigenvalues (or characteristic roots) of a matrix  $\mathbf{A}$ . The equation  $|\mathbf{A} - \lambda\mathbf{I}| = 0$  (as a function of  $\lambda$ ) is called the characteristic equation.  $Ax = \lambda x$

- The eigenvector associated with  $\lambda_i$  is the solution  $\mathbf{x}$  to  $(\mathbf{A} - \lambda_i\mathbf{I})\mathbf{x} = \mathbf{0}$ . It is usual practice to determine an eigenvector so that it has length unity.

$$Ax_i = \lambda_i x_i : \det(\mathbf{A} - \lambda\mathbf{I}) = 0. = \det \begin{pmatrix} a_{11}-\lambda & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22}-\lambda & & \\ \vdots & & \ddots & \\ a_{k1} & & & a_{kk}-\lambda \end{pmatrix} = C_k \lambda^k + C_{k-1} \lambda^{k-1} + \dots + C_1 \lambda + C_0 = 0$$

$C_i$  depends on entries of  $\mathbf{A}$ .

Solve  $(\mathbf{A} - \lambda_i \mathbf{I}_k) \mathbf{x} = \mathbf{0}$  to find  $\mathbf{x}_i$

if  $\det \neq 0 \Rightarrow$  inverse  $\Rightarrow \mathbf{x} = (\mathbf{A} - \lambda_i \mathbf{I}_k)^{-1} \cdot \mathbf{0}$

$$\lambda^2 - 4\lambda + 4 = 0$$

$$\lambda_1 = \lambda_2 = 2$$

multiple eigenvalue

$Ax = \lambda x \ (x \neq 0)$   
why eigenvalues solve  $\det(A - \lambda I) = 0$ .

$$Ax = \lambda x \Leftrightarrow (A - \lambda I)x = 0$$

this means  $\Rightarrow X_1 \cdot \text{1col of } A - \lambda I + X_2 \cdot \text{2col of } A - \lambda I + \dots + X_k \cdot \text{kcol of } A - \lambda I$

$\stackrel{=0.}{\searrow} \rightarrow$  linear dependent  
 $(x \neq 0) \Leftrightarrow \text{rank}(A - \lambda I) < k$

$$\Leftrightarrow \det(A - \lambda I) = 0.$$

# Eigenvalue decomposition

$$\begin{aligned}
 A^{k \times k} &= U \Lambda U^T = (u_1, u_2, \dots, u_k) \overset{\text{orthogonal}}{\begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_k \end{pmatrix}} \begin{pmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_k^T \end{pmatrix} \\
 &= (\lambda_1 u_1, \lambda_2 u_2, \dots, \lambda_k u_k) \begin{pmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_k^T \end{pmatrix} \\
 &= \sum_{i=1}^k \lambda_i u_i u_i^T
 \end{aligned}$$

usually  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$

Theorem [eigenvalue decomposition of symmetric matrix]

Let  $\mathbf{A}$  be a  $k \times k$  square symmetric matrix. Then  $\mathbf{A}$  can be expressed in terms of its  $k$  eigenvalue-eigenvector pairs  $(\lambda_i, \mathbf{u}_i)$  as

If  $\lambda_i \neq \lambda_j$ , then  $u_i^T u_j = 0$ .

$$\begin{cases} A u_i = \lambda_i u_i \rightarrow u_j^T A u_i = u_j^T \lambda_i u_i \\ A u_j = \lambda_j u_j \rightarrow u_i^T A u_j = u_i^T \lambda_j u_j \end{cases}$$

might  $\lambda_i = 0$ , might  $\lambda_i = \lambda_j$  where  $i < j$

$$\mathbf{A} = \sum_{i=1}^k \lambda_i \mathbf{u}_i \mathbf{u}_i^T.$$

In a matrix form, we can write  $\mathbf{A} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T$  where  $\mathbf{U}$  is an  $k \times k$  orthogonal matrix whose  $i$ th column is the eigenvector  $\mathbf{u}_i$  of  $\mathbf{A}$ , and  $\mathbf{\Lambda}$  is the diagonal matrix whose diagonal elements are the corresponding eigenvalues, and usually sorted in descending order.

why we have such decomposition:

(For each distinct eigenvalue find associated eigenvector

(For multiple eigenvalue, Gram-Schmidt to get orthogonal basis  $\mathcal{B}_U = (A u_1, A u_2, \dots, A u_k) = (\lambda_1 u_1, \lambda_2 u_2, \dots, \lambda_k u_k)$

# Example

For example, let

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = U \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} U^T$$

where

$$U = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

orthogonal transform in 2-dimensional case

$$A = \begin{bmatrix} 2.2 & .4 \\ .4 & 2.8 \end{bmatrix}$$

$$A = A I = U \Lambda U^T = U \Lambda U^T$$

$$= (u_1, u_2, \dots, u_k) \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_k \end{bmatrix}$$

$$u_1? (A - \lambda_1 I_2) x = 0$$

Solve  $x$

$$\text{If } x_1 \text{ solves } (A - \lambda_1 I_2) x_1 = 0$$

$$\text{So } 2x_1 \text{ solves } (A - \lambda_1 I_2) x_1 = 0$$

$$\text{So make } x_1 \text{ has unit length}$$

Then

$$|A - \lambda I| = \lambda^2 - 5\lambda + 6.16 - .16 = (\lambda - 3)(\lambda - 2)$$

$u_1 = \frac{x_1}{\|x_1\|_2}$

so  $A$  has eigenvalues  $\lambda_1 = 3$  and  $\lambda_2 = 2$ . The corresponding eigenvectors are  $u_1^T = [1/\sqrt{5}, 2/\sqrt{5}]$  and  $u_2^T = [2/\sqrt{5}, -1/\sqrt{5}]$ , respectively. Consequently,   
check that  $u_1^T u_2 = 0$   
 $u_2^T u_1 = 0$   
not unique as  $-u_1^T$  solve the equation.

$$A = \begin{bmatrix} 2.2 & .4 \\ .4 & 2.8 \end{bmatrix} = 3 \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} + 2 \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{-1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} \end{bmatrix}.$$

- The spectral decomposition is an important analytical tool. With the eigen-decomposition of sample covariance matrices, we are very easily able to demonstrate certain statistical results (later in this course).

- Another advantage is certain matrix operation becomes easier, e.g., calculating  $A^{-1}$  or  $A^{100}$ . We can also find the square root of a positive-definite matrix using eigenvalue decomposition.

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# Singular Value Decomposition

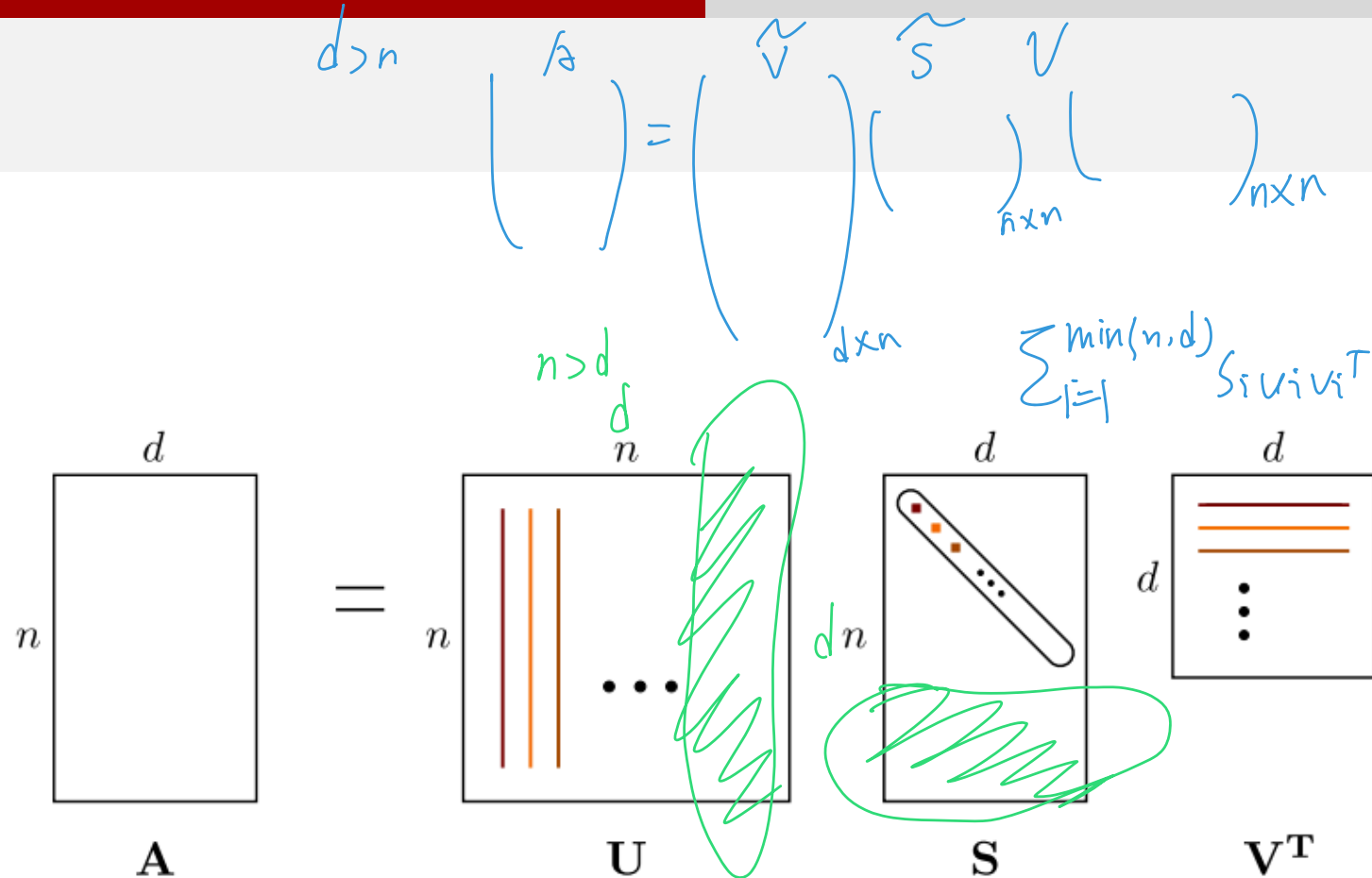
A *Singular Value Decomposition* (SVD) of an  $n \times d$  matrix  $\mathbf{A}$  expresses the matrix as the product of three “simple” matrices:

$$\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$$

where:

- $\mathbf{U}$  is an  $n \times n$  orthogonal matrix
- $\mathbf{V}$  is an  $d \times d$  orthogonal matrix;
- $\mathbf{S}$  is an  $n \times d$  matrix with non-zero entries on main-diagonal, sorted in descending order.

## SVD



**Figure:** The singular value decomposition (SVD). Each singular value in  $\mathbf{S}$  has an associated left singular vector in  $\mathbf{U}$ , and right singular vector in  $\mathbf{V}$ .

## SVD

$$\mathbf{A} = \mathbf{U} \mathbf{S} \mathbf{V}^T \quad \mathbf{A} \in \mathbb{R}^{n \times d}$$

$$\mathbf{A} \mathbf{A}^T = \mathbf{U} \mathbf{S} \mathbf{U}^T \mathbf{U} \mathbf{S} \mathbf{V}^T \mathbf{V}^T$$

$$= \mathbf{U} \mathbf{S} \mathbf{S}^T \mathbf{U}^T$$

$$= \sum_{i=1}^n s_i^2 \mathbf{u}_i \mathbf{u}_i^T$$

eigen decomposition

$$s_i = \sqrt{\lambda_i} \text{ where } \lambda_i \text{ are non-zero eigenvalues of } \mathbf{A} \mathbf{A}^T$$

$$\mathbf{u}_i \text{ are eigenvectors.}$$

$$\mathbf{X} = \sum_{i=1}^{320} s_i \mathbf{u}_i \mathbf{v}_i^T \quad s_1 = 6, s_2 = 5, s_{12} = 1, s_n = 0 \dots$$

$$s_1 \geq s_2 \geq \dots \geq s_{320}$$

$$\mathbf{X}^T = \sum_{i=1}^5 s_i \mathbf{v}_i \mathbf{u}_i^T$$

$$5 \times (n+1) \leq n \times d$$

- The columns of  $\mathbf{U}$  and  $\mathbf{V}$  are called the *left singular vectors* and *right singular vectors* of  $\mathbf{A}$ , respectively. The entries on the main-diagonal position are *singular values* of  $\mathbf{A}$ .
  - Another way to write the SVD is  $\mathbf{A} = \sum_{i=1}^{\min\{n,d\}} s_i \mathbf{u}_i \mathbf{v}_i^T$ , where  $\mathbf{u}_i$ ,  $\mathbf{v}_i$  are the  $i$ -th column of  $\mathbf{U}$ ,  $\mathbf{V}$ , respectively.
  - How to obtain SVD of  $\mathbf{A}$ ? The columns of  $\mathbf{V}$  are the eigenvectors of  $\mathbf{A}^T \mathbf{A}$ , the singular values are the square roots of the eigenvalues of  $\mathbf{A}^T \mathbf{A}$ , and the columns of  $\mathbf{U}$  are the eigenvectors of  $\mathbf{A} \mathbf{A}^T$ .
- Remark: When  $n > d$ , computing those eigenvectors associated with non-zero eigenvalues of  $\mathbf{A} \mathbf{A}^T$  is sufficient to obtain the SVD.

$$\mathbf{A}^T \mathbf{A} = \mathbf{V} \mathbf{S}^T \mathbf{U}^T \mathbf{U} \mathbf{S} \mathbf{V}^T$$

$$= \mathbf{V} \mathbf{S}^T \mathbf{S} \mathbf{V}^T$$

$$\begin{pmatrix} \dots & 0 & \dots \\ \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \end{pmatrix} \begin{pmatrix} \dots & 0 & \dots \\ \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \end{pmatrix} \Rightarrow \begin{bmatrix} s_1^2 & & & \\ & s_2^2 & & \\ & & \ddots & \\ 0 & & & s_{\min(n,d)}^2 \\ & & & & 0 \end{bmatrix}$$

# Positive definite matrix and quadratic form

## Positive-definite matrix & quadratic form

When a  $k \times k$  symmetric matrix  $\mathbf{A}$  satisfies

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$$

for all  $\mathbf{x}^\top = [x_1, x_2, \dots, x_k] \in \mathbb{R}^k$ , the matrix  $\mathbf{A}$  is said to be *positive semi-definite* (or *nonnegative definite*), denoted by  $\mathbf{A} \succeq 0$ . If equality holds only for the vector  $\mathbf{x}^\top = [0, 0, \dots, 0]$ , then  $\mathbf{A}$  is said to be *positive definite*, denoted by  $\mathbf{A} \succ 0$ .

Because  $\mathbf{x}^\top \mathbf{A} \mathbf{x}$  has only squared terms  $x_i^2$  and product terms  $x_i x_k$ , it is called a *quadratic form*. If  $\mathbf{A}$  is positive definite,  $\mathbf{x}^\top \mathbf{A} \mathbf{x}$  is called a positive definite quadratic form.

A  $k \times k$  symmetric matrix  $\mathbf{A}$  is a positive definite matrix if and only if every eigenvalue of  $\mathbf{A}$  is positive!

# Quadratic form and Distance

If we use the distance formula introduced in Chapter 1, the distance from the origin satisfies the general formula

$$(\text{distance})^2 = a_{11}x_1^2 + a_{22}x_2^2 + \cdots + a_{pp}x_p^2 + 2(a_{12}x_1x_2 + a_{13}x_1x_3 + \cdots + a_{p-1,p}x_{p-1}x_p)$$

provided that  $(\text{distance})^2 > 0$  for all  $[x_1, x_2, \dots, x_p] \neq [0, 0, \dots, 0]$ .

Setting  $a_{ij} = a_{ji}$ ,  $i \neq j$ ,  $i = 1, 2, \dots, p$ ,  $j = 1, 2, \dots, p$ , we have

$$0 < (\text{distance})^2 = [x_1, x_2, \dots, x_p] \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ a_{21} & a_{22} & \cdots & a_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pp} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$$

or

$$0 < (\text{distance})^2 = \mathbf{x}^\top \mathbf{A} \mathbf{x} \quad \text{for } \mathbf{x} \neq \mathbf{0}.$$

$A A^T$

For any matrix  $A \in \mathbb{R}^{n \times n}$

$$A A^T \succeq 0$$

For any  $x \in \mathbb{R}^n$

$$x^T A A^T x = (A^T x)^T (A^T x) \geq 0$$

$$= y^T y = \|y\|_2^2 \geq 0$$

$\geq 0$

# Geometrical understanding

- Expressing distance as the square root of a positive definite quadratic form allows us to give a geometrical interpretation based on the eigenvalues and eigenvectors of the matrix  $\mathbf{A}$ .
- For example, suppose  $p = 2$ . Then the points  $\mathbf{x}^\top = [x_1, x_2]$  of constant distance  $c$  from the origin satisfy

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = a_{11}x_1^2 + a_{22}x_2^2 + 2a_{12}x_1x_2 = c^2$$

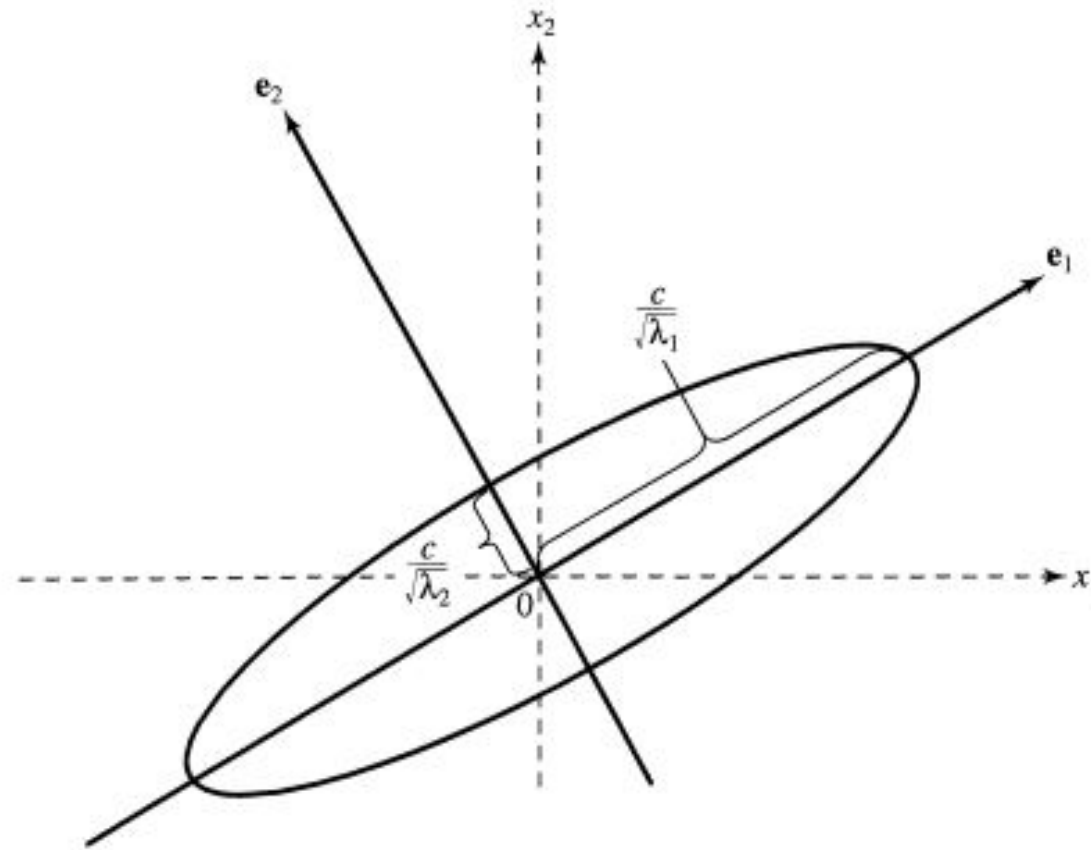
By the spectral decomposition,

$$\mathbf{A} = \lambda_1 \mathbf{e}_1 \mathbf{e}_1^\top + \lambda_2 \mathbf{e}_2 \mathbf{e}_2^\top \text{ so } \mathbf{x}^\top \mathbf{A} \mathbf{x} = \lambda_1 \left( \mathbf{x}^\top \mathbf{e}_1 \right)^2 + \lambda_2 \left( \mathbf{x}^\top \mathbf{e}_2 \right)^2$$

Now,  $c^2 = \lambda_1 y_1^2 + \lambda_2 y_2^2$  is an ellipse in  $y_1 = \mathbf{x}^\top \mathbf{e}_1$  and  $y_2 = \mathbf{x}^\top \mathbf{e}_2$  because  $\lambda_1, \lambda_2 > 0$  when  $\mathbf{A}$  is positive definite.

# Geometrical understanding

- We easily verify that  $\mathbf{x} = c\lambda_1^{-1/2}\mathbf{e}_1$  satisfies  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \lambda_1 \left( c\lambda_1^{-1/2}\mathbf{e}_1^\top \mathbf{e}_1 \right)^2 = c^2$ . Similarly,  $\mathbf{x} = c\lambda_2^{-1/2}\mathbf{e}_2$  gives the appropriate distance in the  $\mathbf{e}_2$  direction.



$A \succeq 0$ : given  $a \geq 0$ ,  $b^2 = a$ , in matrix case:  $B^2 = A$ , such  $B$  is the square root of  $A \succeq 0$ . Problem:  $B$ ? Solution:  $A = U\Lambda U^T$ . Let  $B = U\Lambda^{1/2}U^T$ , thus  $B^2 = U\Lambda^{1/2}U^T U\Lambda^{1/2}U^T = U\Lambda^{1/2}\Lambda^{1/2}U^T = U\Lambda U^T$  since  $U^T U = I_n$ . Problem is this  $B$  unique? if  $B_2 = -B$ , then  $B_2^2 = (-B)^2 = A$ . Or usually we use the positive semi-definite square root.

## Square-root matrix

$B_3 = (U_1, \dots, U_n) \begin{pmatrix} \sqrt{\lambda_1} & & \\ & -\sqrt{\lambda_2} & \\ & & \sqrt{\lambda_3} \\ & & & \ddots \\ & & & & \sqrt{\lambda_n} \end{pmatrix} \begin{pmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_n^T \end{pmatrix}$   $B_3 = A$  as well so  $\begin{cases} B = U\Lambda^{1/2}U^T \\ B \succeq 0 \end{cases} \Rightarrow B$  is unique.

The square-root matrix, of a positive semi-definite matrix  $A$ ,

$$A^{1/2} = \sum_{i=1}^k \sqrt{\lambda_i} \mathbf{u}_i \mathbf{u}_i^T = \mathbf{U} \Lambda^{1/2} \mathbf{U}^T$$

has the following properties:

- $(A^{1/2})^T = A^{1/2}$  (that is,  $A^{1/2}$  is symmetric).
- $\underbrace{A^{1/2} A^{1/2}} = A$ .
- $(A^{1/2})^{-1} = \sum_{i=1}^k \frac{1}{\sqrt{\lambda_i}} \mathbf{u}_i \mathbf{u}_i^T = \mathbf{U} \Lambda^{-1/2} \mathbf{U}^T$ , where  $\Lambda^{-1/2}$  is a diagonal matrix with  $1/\sqrt{\lambda_i}$  as the  $i$ th diagonal element.
- $A^{1/2} A^{-1/2} = A^{-1/2} A^{1/2} = \mathbf{I}$ , and  $A^{-1/2} A^{-1/2} = A^{-1}$ , where  $A^{-1/2} = (A^{1/2})^{-1}$ .

# Trace

Let  $\mathbf{A} = \{a_{ij}\}$  be a  $k \times k$  square matrix. The *trace* of the matrix  $\mathbf{A}$ , written  $\text{tr}(\mathbf{A})$ , is the sum of the diagonal elements; that is,  $\text{tr}(\mathbf{A}) = \sum_{i=1}^k a_{ii}$ . Let  $\mathbf{A}$  and  $\mathbf{B}$  be  $k \times k$  matrices and  $c$  be a scalar. *have it so generally w.*

- $\text{tr}(c\mathbf{A}) = c\text{tr}(\mathbf{A})$

- $\text{tr}(\mathbf{A} \pm \mathbf{B}) = \text{tr}(\mathbf{A}) \pm \text{tr}(\mathbf{B})$

*$\text{tr}(\mathbf{A}\mathbf{B}\mathbf{C}) \neq \text{tr}(\mathbf{C}\mathbf{B}\mathbf{A})$  because  $\text{tr}(\mathbf{A}\mathbf{B}\mathbf{C}) = \text{tr}(\mathbf{B}\mathbf{C}\mathbf{A})$*

- $\text{tr}(\mathbf{A}\mathbf{B}) = \text{tr}(\mathbf{B}\mathbf{A}) \Rightarrow \text{tr}(\mathbf{A}_1\mathbf{A}_2\mathbf{A}_3 \cdots \mathbf{A}_{k-1}\mathbf{A}_k) = \text{tr}(\mathbf{A}_k\mathbf{A}_1\mathbf{A}_2 \cdots \mathbf{A}_{k-1}) = \text{tr}(\mathbf{A}_{k-1}\mathbf{A}_k \cdots \mathbf{A}_{k-2})$

- $\text{tr}(\mathbf{B}^{-1}\mathbf{A}\mathbf{B}) = \text{tr}(\mathbf{A})$

*$\text{tr}(\mathbf{A}\mathbf{A}^T) = \text{tr}(\mathbf{A}^T\mathbf{A}) = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2, \mathbf{A} \in \mathbb{R}^{m \times n}$*

- $\text{tr}(\mathbf{A}\mathbf{A}^T) = \sum_{i=1}^k \sum_{j=1}^k a_{ij}^2$

*Frobenius norm:  $\|\mathbf{A}\|_F = \sqrt{\text{tr}(\mathbf{A}\mathbf{A}^T)} = \sqrt{\sum \sum a_{ij}^2}$ , F-norm.*

*Spectral norm:  $\|\mathbf{A}\|_2 = \sup_{\|x\|=1} \|\mathbf{A}x\|$  for  $\mathbf{A} \in \mathbb{R}^{m \times n}$  = largest singular value of  $\mathbf{A}$ .*

- If  $\mathbf{A}$  is symmetric,  $\text{tr}(\mathbf{A}) = \sum_{i=1}^k \lambda_i$

*↗ singular values.*

*$\sum_{i=1}^r \|u_i\|^2 = r$   
 $\|\mathbf{A} - \mathbf{B}\|_F$   
 $\text{rank}(\mathbf{B}) = r$*

*$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^{\min(m,n)} \sigma_i^2}$*

Chapter 2

